Ten-Layer Source-to-Support Review

RH-140–RH-149 and an Inclusion-Minimal Three-Interface Frontier

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Abstract

We review RH-140–RH-149 as one interval-to-directional-support proof system. The ten layers sharpen normalized source perturbation, isolate spectral packet gaps, close all ten frozen source packets with factorized Arb arithmetic, certify strict threshold branches, compute exact backward viability kernels, separate finite bad births from recurrent obstructions, derive a sharp projective normalized-base recurrence, replace its highly lossy unsigned product by a correlated support cocycle, and state the complete conditional source-to-support composition theorem.

The finite progression is strong: 10/10 factorized rank-four packets, 360/360 strict nominal threshold margins, 330/330 outward residual pairs, 28/30 common positive support tubes, and an 18/18 delayed clean suffix. The main theoretical conclusion is nevertheless conditional. Within the current modular architecture, eventual positive directional support follows from exactly three interfaces:

$$E_{\text{source}}, \qquad E_{\text{update}}, \qquad E_{\text{cocycle}}.$$

They respectively provide all-level source/packet enclosures, branch-stable transport into the outward tail/base assembly, and a delayed correlated support cocycle with bounded logarithmic drawdown. This set is inclusion-minimal: RH-148 supplies sufficiency, while a source-identification, branch/outward, or cocycle omission witness defeats each proper two-interface subset.

The route is also revised. Summable Hilbert projective variation remains a rigorous sufficient fallback, but its finite chain totals range from 97.5 to 234.2 and do not support summability. Direct correlated reanchoring recovers 16–48 orders of magnitude, so the support cocycle becomes the primary asymptotic route. The next priority is E_{update}: propagate the RH-142 packet balls through every RH-143 update into the RH-138 outward assembly. All nine upstream archives pass review and no Stage A, Hilbert–Polya, or Riemann-Hypothesis claim is asserted.

1 The RH-139 frontier revisited

RH-139 ended the previous ten-layer block with four packets. For exact matrices it required controlled viability P_V and a positive normalized base P_B; connection to the intended model added source enclosure P_E and, for independent assembly, outward radii P_O.

RH	type	layer	exact or finite consequence
140	constructive	normalized snapshot	sharp operator/Frobenius/trace bounds; 10 rank-four radii below 10^{-3}
141	mixed	packet perturbation	sharp gap gate; 4/10 universal packets and 10/10 quadratic diagnostics
142	constructive	factorized Arb closure	10/10 frozen binary packets, with two interval-eigenvalue rescues
143	constructive	threshold branch	exact clipped radius; 360/360 strict local margins
144	mixed	backward viability	28 full kernels and two candidate-family obstructions
145	mixed	delayed start	finite-prefix invariance; 18/18 clean suffix from $\sigma = 0.04$
146	mixed	projective base	sharp recurrence; finite projective product is extremely lossy
147	constructive	correlated tube	signed support cocycle; 28 common 10^{-10} tubes
148	synthesis	conditional composition	source, branch, outward, and cocycle gates imply support
149	synthesis	frontier review	inclusion-minimal three-interface route and revised priorities

Table 1: The RH-140–RH-149 proof stack.

RH-140–RH-149 do not magically prove those all-level statements. They change their organization. First, RH-140–RH-143 split P_E into source, packet, and branch-stability interfaces. Second, RH-144–RH-145 show that finite viability should be handled by backward kernels and delayed start, not per-step contraction. Third, RH-146–RH-147 show that P_V and P_B should be transported together as one directional-support cocycle. Finally, RH-148 absorbs P_O into the exact update-to-outward interface. These reorganizations use standard perturbation and interval principles [Kato, 1995, Bhatia, 2007, Moore, 1966] together with viability theory [Aubin, 1991].

The revised map is

$$(P_E, P_O) \longmapsto (E_{\text{source}}, E_{\text{update}}), \qquad (P_V, P_B) \longmapsto E_{\text{cocycle}}.$$

The count drops from four packets to three interfaces because two pairs are now composed rather than independently optimized.

2 What the ten layers established

Table 1 records the role of each paper. “Mixed” means that a constructive theorem and a sharp wall are both central.

The enclosure side improved qualitatively. RH-141’s universal snapshot ball certified only four rank-four packets; RH-142 retained the source factorization and used interval eigenvalues to certify all ten. RH-143 then proved that all 360 nominal threshold choices are strictly separated from their branch walls. The crucial claim boundary remains: the RH-142 packet balls have not yet been propagated through those recursive updates.

The viability side also changed. The two failed coarse chains are not greedy artifacts: RH-144 proves every available control has zero-state floor above one. RH-145 shows they occur at one finite anchor and that deleting this prefix leaves 18 clean chains. This makes delayed start logically valid but does not exclude recurrence at future untested levels.

3 An inclusion-minimal frontier

We formalize the revised interface set.

Theorem 1 (Three-interface support frontier)**.** *Within the RH-148 modular architecture, the set*

$$\mathcal{E} = \{E_{\mathrm{source}}, E_{\mathrm{update}}, E_{\mathrm{cocycle}}\}$$

is sufficient and inclusion-minimal for eventual positive directional support.

Here E_{source} certifies the exact normalized source and spectral packet at every relevant level; E_{update} preserves the exact threshold branch and supplies outward tail maps and base-ratio lowers on that same construction; and E_{cocycle} supplies a delayed reset and a uniform lower bound on all partial logarithmic support sums.

Proof. Sufficiency is the source-to-support composition theorem of RH-148: the three interfaces imply

$$B_n \geq B_N e^{-C} > 0$$

after the delayed start.

For inclusion-minimality, omit E_{source}. A nominal eigenspace may then be unrelated to the exact packet at a multiple or unresolved eigenvalue, so later bounds can apply to the wrong object. Omit E_{update}. A threshold contact can change the branch, or a nominal tail recurrence can underestimate an actual unit tail, making the actual support zero. Omit E_{cocycle}. The admissible equality multipliers $\kappa_n = 1/2$ drive support to zero. Thus every proper two-interface subset has a witness for which the remaining interfaces hold only nominally or locally while the conclusion fails. □

The minimality is architecture relative. A future analytic construction may prove two interfaces simultaneously or bypass a finite packet representation, but it must still resolve source identification, exact update/outward transport, and long-run support loss in some form.

An automated truth-table audit evaluates all eight subsets of the three interfaces. Exactly one—the full set—is marked closed, and all three single omissions are linked to explicit RH-148 witnesses.

4 Primary and fallback routes

RH-146 supplies a clean fallback:

$$\sum_n d_{\mathrm{proj}}(O_n^T G_n O_n, G_{n+1}) < \infty \quad \Longrightarrow \quad \liminf a_n > 0.$$

The theorem is sharp, but the frozen physical-gauge audit is unfavorable. Step distances never fall below 3.92, total chain variation lies between 97.53 and 234.16, and the terminal

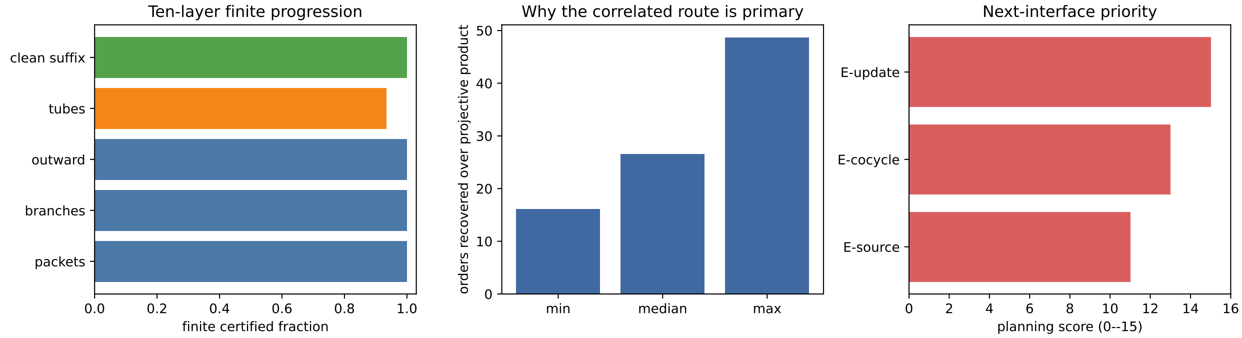

Figure 1: Finite progression, orders recovered by correlated reanchoring, and the revised interface priorities.

universal lowers fall as low as 1.66×10^{-52}. The observed sequence is too short to disprove a different asymptotic normalization, but it gives no positive evidence for direct summability.

RH-147's signed support cocycle retains both losses and recoveries. Its statewise floors exceed the projective products by 16.13–48.65 orders, with median recovery 26.54 orders. Ninety-eight of 292 positive local support ratios exceed one. This is why the correlated route becomes primary: not because it is already proved all levels, but because it preserves structure that the fallback deliberately discards.

5 Priority of the next interface

The open interfaces are not equally mature. We use a transparent planning score—tractability, falsification value, and dependency leverage, each on a zero-to-five scale. This is a research-management heuristic, not a theorem. It ranks

$$E_{\text{update}} : 15, \qquad E_{\text{cocycle}} : 13, \qquad E_{\text{source}} : 11.$$

E_{update} comes first because both endpoints already exist. RH-142 supplies ten certified packet balls and RH-143 supplies exact local branch margins; RH-138 supplies the target outward residual architecture. Transport through the middle is finite, concrete, and highly falsifiable. If an interval radius crosses a branch wall or destroys a Gram lower, the failure identifies the precise level and mechanism before a harder asymptotic theory is attempted.

The proposed next block is therefore:

1. propagate all RH-142 packet balls through every thresholded recursive update, with outward branch radii;

2. rebuild the Gram/tail assembly from those transported intervals and certify the RH-138 residuals on the same objects;

3. apply RH-148 to obtain, or refute, the first finite end-to-end source-to-support interval certificate on the delayed clean suffix;

4. only then refine anchors and measure the scale laws needed for E_{source} and E_{cocycle}.

A negative result at either of the first two steps is informative. It may show that the packet gauge, threshold rule, depth, or source normalization must change. Such a wall should be recorded rather than hidden by resetting to nominal matrices.

6 Archive review and claim boundary

The automated review hashes the nine RH-140–RH-148 summaries, verifies all nine archive records, checks the previous RH-139 frontier, and scans every program boundary. There are zero Stage A, Hilbert–Polya, or RH flags. The finite headline remains:

$$10/10 \text{ packets}, \quad 360/360 \text{ branches}, \quad 330/330 \text{ outward residual pairs},$$

$$28/30 \text{ positive tubes}, \quad 18/18 \text{ clean delayed-suffix chains}.$$

These ratios belong to separate, carefully bounded finite audits. Until E_{update} is closed, they are not one interval end-to-end result.

RH-149 proves an inclusion-minimal conditional frontier and revises the route priority. It does not close the finite interval bridge, prove any of the three interfaces all levels, establish uniform Stage A, construct a Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

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